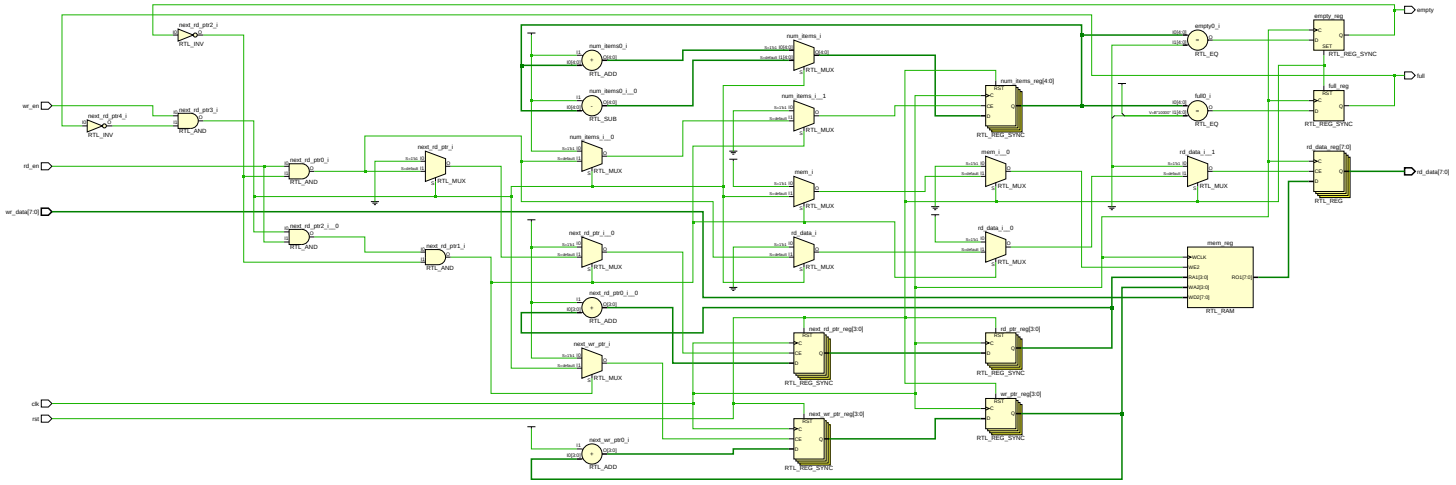

wr_en
rd_en
wr_data[7:0]
clk
rst
empty
full
rd_data[7:0]
RTL_INV
RTL_AND
RTL_MUX
RTL_ADD
RTL_SUB
RTL_EQ
RTL_REG_SYNC
RTL_REG
RTL_RAM
num_items_i
num_items_reg[4:0]
mem_i
mem_reg
rd_data_i
rd_data_reg[7:0]
empty_reg
full_reg
next_rd_ptr_i
next_rd_ptr_reg[3:0]
next_wr_ptr_i
next_wr_ptr_reg[3:0]
rd_ptr_reg[3:0]
wr_ptr_reg[3:0]